



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA1017 – SOFTWARE ENGINEERING

INDIVIDUAL REFLECTION REPORT

INTELLIGENT DISASTER EARLY WARNING AND EVACUATION MANAGEMENT SYSTEM

NAME: OSURU SIDDI RISHWANTH

REGISTER NO.: 192472148

ROLE: Alerts, Citizen Portal, and Communication Workflow

DESIGN / DEVELOPMENT DECISION: I contributed to the alert-composer workflow, citizen portal, public guidance, and incident-reporting process. I supported the decision to require hazard type, severity, affected area, message, expiry, and communication channels before an alert preview can be created.

CHALLENGE FACED AND SOLUTION: The main challenge was balancing urgency with accuracy. Emergency messages must be easy to understand but should not be mistaken for an official order when they are produced by a prototype. I addressed this through clear severity labels, visible demo notices, human-review wording, and separate authority and citizen experiences.

LEARNING GAINED: I learned about communication design, form validation, accessibility, plain-language writing, and the responsibility involved in displaying safety information. I also learned that a citizen reporting feature must collect useful location and severity details while avoiding unnecessary personal information.

SDG RELEVANCE: The project supports SDG 11 by improving community access to understandable safety information and participation in incident reporting.

COURSE OUTCOME CONTRIBUTION: My contribution supports the course outcome related to web development, communication design, and human-computer interaction through the alert, guidance, and citizen-reporting workflows.

POSSIBLE IMPROVEMENT: Future improvements include multilingual messages, offline access, official SMS and mobile-notification integration, and user testing with diverse community groups.